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A CONSTRUCTING TRANSFORMERS THAT UTILIZE THE CAPACITY OF THE EMBEDDING SPACE

A.1 UNCONDITIONAL GENERATION FOR A SINGLE VIRTUAL TOKEN

This section provides an explicit construction of a transformer with the properties described in Theorem 1. The goal is to construct a transformer that, by varying the choice of the virtual token, can generate any sequence of N tokens.

First, we need to specify how we encode the target sequence $(Y_1, \dots, Y_N)$ into the virtual token s_1 . We chose the size of the embedding (and hence of s_1) to be N . This way, each element of s_1 can represent one position of the target sequence. We then represent the token value by discretizing each element of s_1 into V levels:

$$s_1 = ((Y_1-1)/V, \dots, (Y_N-1)/V).$$

Note that this means that each element of s_1 is in $[0, 1)$.

When predicting the token for the $i + 1$ position, the transformer needs to pick the i -th element of s_1 , and then decode the corresponding value as a one-hot encoding representing the Y_i -th token.

We extract the i -th element of s_1 using one attention block of two heads. The `fst` head always looks at the first position which is our virtual token s_1 . For that purpose we create an attention head that always has $A_{ij}^{\text{fst}} = 1$ if $j = 1$ and $A_{ij}^{\text{fst}} = 0$ otherwise together with a value matrix W_V^{fst} that extracts the embedding. This is achieved with

$$W_Q^{\text{fst}} = [0_N, \mathbf{1}_N], \quad W_K^{\text{fst}} = [0_N, 1, \mathbf{1}_{N-1}], \quad W_V^{\text{fst}} = [I_N, 0_{N \times N}], \quad (8)$$

and a sufficiently high inverse temperature parameter T .

The `pos` head instead extracts the one-hot encoding of the current position. This can be done with an attention head that always attends only to the current position and a value matrix W_V^{pos} that extracts the position embedding as a one-hot vector:

$$W_Q^{\text{pos}} = [0_{N \times N}, I_N], \quad W_K^{\text{pos}} = [0_{N \times N}, I_N], \quad W_V^{\text{pos}} = [0_{N \times N}, I_N]. \quad (9)$$

When the outputs of these two attention heads are summed, then only the element of s_1 that corresponds to the current position will be larger than 1. From Equation (2) the output at the i -th position of the attention block is:

$$t_i = \sum_{j=1}^p A_{ij}^{\text{fst}} x_j + \sum_{j=1}^p A_{ij}^{\text{pos}} e_N(j) = s_1 + e_N(i),$$

where $x_1 = s_1$ and $x_j = E_{:, Y_{j-1}}$ for $j > 1$.

We can extract the value of s_1 corresponding to the current position by subtracting 1 from the hidden state and apply ReLU: $\hat{\mathcal{L}}_{\text{ex}} = \hat{\mathcal{L}}[I_N, -\mathbf{1}_N]$. Now, we are left with only one non-zero entry and that's the one corresponding to the next token. We can retain only the non-zero entry if we just sum all the entries of the hidden state with $\hat{\mathcal{L}}_{\text{sum}} = \hat{\mathcal{L}}[\mathbf{1}_N^\top, 0]$.

The final step is to map this scalar to a V -dimensional vector which has its maximum value at index Y_i . This task is equivalent to designing V linear functions, each attaining its maximum at one of $0, 1/V, \dots, (V-1)/V$. To construct this, we use the property of convex functions that their tangent is always under the plot of the function. Therefore, given a convex function $\gamma(x)$, we construct the i -th linear function to be simply the tangent of γ at i^{-1}/V . If we take $\gamma(x) = (x - 1/2)^2$, this results in the following linear layer:

$$\mathcal{L}_{\text{proj}} = \mathcal{L} \left[\left[\frac{2(1-1)}{V} - 1, \dots, \frac{2(V-1)}{V} - 1 \right]^T, \left[\frac{1}{4} - \frac{(1-1)^2}{V^2}, \dots, \frac{1}{4} - \frac{(V-1)^2}{V^2} \right]^T \right]. \quad (10)$$

Figure 4 shows the predictors for each individual token id.

With just two attention heads and three linear layers, the transformer $\mathcal{A}[(W_Q^{\text{fst}}, W_Q^{\text{pos}}), (W_K^{\text{fst}}, W_K^{\text{pos}}), (W_V^{\text{fst}}, W_V^{\text{pos}})] \circ \hat{\mathcal{L}}_{\text{ex}} \circ \hat{\mathcal{L}}_{\text{sum}} \circ \mathcal{L}_{\text{proj}} \circ \text{softmax}$ achieves the upper bound of V^N unique outputs by controlling a single virtual token at its input. Note that for this